AE 462
Lab 3
Lab Session: Monday, February 23, 2026
Due: Monday, March 2, 2026, 9am CST

This laboratory assignment will focus on running CFD simulations in parallel and performing validation through comparison to experimental data. The items required for your assignment submission are given here, alongside helpful instructions for completing this lab. Provide your responses in a single PDF file. Upload any scripts, spreadsheets, or other work used during data analysis.

Report Instructions:
Using CFD-generated data from this exercise, provide a response to the following:

1) **Parallel computing time:**
   a. Plot the convergence time of your 250-iteration processing time vs. number of logical processes specified when calling SU2 (i.e., NP from prior instructions). Provide the execution time, in seconds, on the *y* axis and the number of logical processes on the *x* axis. Label your axes and use units where applicable.
   b. Based on your observations of the execution time, describe the benefits and challenges of parallel computing in CFD in 2-3 sentences. In particular, indicate any observed differences between what happens when doubling logical processes with limited computational demand (e.g., mpiexec -n 1 to mpiexec -n 2) versus that produced when doubling across a greater number of logical processes (e.g., mpiexec -n 4 to mpiexec -n 8).
2) **Comparison of results to experimental data:**
   For this case, use the experimental airfoil performance data of Ladson et al. and Gregory and O'Reilly, which is linked on the NASA Turbulence Modeling Resource page: https://turbmodels.larc.nasa.gov/naca0012_val.html
   a. Co-plot your CFD predictions of the airfoil C_l and C_d at $\alpha = 0$, 10, and 15 deg with the experimental data from Ladson et al. at $Re_c = 6 \times 10^6$. For your CFD simulations, you can find these variables stored in the corresponding "history.csv" file. Provide two plots, one with C_l on the *y* axis and angle of attack (α) on the *x* axis. The other plot should have C_l on the *y* axis and C_d on the *x* axis. Data from Ladson et al. for this case can be found at (https://turbmodels.larc.nasa.gov/NACA0012_validation/CLCD_Ladson_expdata.dat), corresponding to t="80 grit". These data can also be obtained from the course Canvas site for Lab 3, having title "LadsonClCd.csv". Label your axes and use units where applicable. Additional predictions of the airfoil C_l and C_d from other CFD codes are also provided in Table 1 below. Add these data points to your plots as open symbols. Be sure to include a legend and color/symbol representation to sufficiently distinguish these different data sources.
   b. Co-plot your CFD predictions of the airfoil C_p distributions for $\alpha = 10$ deg, with the experimental data of Gregory and O'Reilly, which can be found at (https://turbmodels.larc.nasa.gov/NACA0012_validation/CP_Gregory_expdata.dat) or on the Canvas site for Lab 3, having title "GregoryCp.csv". You can find the C_p distribution data in the "surface_flow" CSV file for your current run. Provide

C_p on the y axis and x/c position on the x axis. Label your axes and use units where applicable. Be sure that your y axis follows the typical convention of external aerodynamics, where smaller (negative, or greater suction) pressure values appear higher on the y axis.

c. Based on your observations, in 2-3 sentences, indicate when and where this current CFD workflow is sufficiently valid. Be sure to include a summary of the underlying assumptions that contribute to this opinion.

Table 1 Aerodynamic performance predictions from NASA CFD packages.

CODE	C_l (α=0 deg)	C_l (α=10 deg)	C_l (α=15 deg)	C_d (α=0 deg)	C_d (α=10 deg)	C_d (α=15 deg)
CFL3D	≈0	1.0909	1.5461	0.00819	0.01231	0.02124
FUN3D	≈0	1.0983	1.5547	0.00812	0.01242	0.02159
NTS	≈0	1.0891	1.5461	0.00813	0.01243	0.02105
JOE	≈0	1.0918	1.5490	0.00812	0.01245	0.02148
SUMB	≈0	1.0904	1.5446	0.00813	0.01233	0.02141
TURNS	≈0	1.1000	1.5642	0.00830	0.01230	0.02140
GGNS	≈0	1.0941	1.5576	0.00817	0.01225	0.02073

Lab Instructions:

Organizing files

Create a new folder within your U:\AE 462 directory called "Lab 3". Download the "naca0012.cfg", "n0012_897-257.su2", and all "restart_flow_AOA.dat" files, where AOA is the angle of attack corresponding to the flow solution. Place these files in your U:\AE 462\Lab 3 directory.

In addition, you will be comparing your CFD results to experiments in this lab exercise. Download the data from Ladson et al. (1987) from the NASA Turbulence Modeling Resource Site, including C_l and C_d data at (https://turbmodels.larc.nasa.gov/NACA0012_validation/CLCD_Ladson_exp), as well as C_p data from Gregory and O'Reilly at (https://turbmodels.larc.nasa.gov/NACA0012_validation/CP_Gregory_expdata.dat).

Timing parallel processing options

For this lab we will be using Windows PowerShell instead of the Command terminal. You can start up the PowerShell terminal directly from your folder. In the Windows Explorer window corresponding to your folder, right click on "Lab 3" and select "**Edit Address**" (or click just to the right of "Lab 3"). Then, type "**powershell**" and press **Enter**. You can also launch PowerShell from the typical search line and use the "cd" command to make your U:\AE 462\Lab 3 folder your active directory.

Your naca0012.cfg file is currently configured to only run 250 iterations of the CFD solver corresponding to the NACA 0012 airfoil at α = 10 deg. With PowerShell open in the correct directory (e.g., "U:\AE 462\Lab 3"), run SU2 with a varying number of processes in parallel to observe the difference in time require to complete the associated runs. You can use PowerShell to time a parallel execution of SU2 with the input (note that the %SU2_RUN% environment variable is not currently configured for PowerShell):

```
Measure-Command {mpiexec -n PR "C:\SU2\v8.4.0-mpi\bin\SU2_CFD" naca0012.cfg}
```

where PR is the number of processes you wish to utilize. When using the Measure-Command function in PowerShell, you will not see the iteration information in the terminal window. However, once the prescribed number of iterations are completed, you will see the time required to run the SU2 case prescribed by the .cfg file.

Run the α = 10 deg case for 250 iterations in parallel using the Measure-Command function in PowerShell for PR = 1, 2, 4, 6, 8, and 10 and record the execution times (in seconds) for later reporting. When running each case, **be patient** as it will typically take 1-2 minutes on a standard EWS machine. If running on a different machine, be sure to identify the CPU information (processor type/generation, clock speed, and number of logical processors). A standard EWS machine is configured with a 12th Generation Intel Core i5-12500, having a base clock speed of 3 GHz and 12 logical processors.

Prepare a warm start

Initializing a CFD run from a previous solution is known as a "warm start" to the current simulation. Open the "naca0012.cfg" file in a text editor. On lines 234 and 247, change "restart_flow" to "**restart_flow_AOA10**". Now, change line 25 to be:

RESTART_SOL= YES

Your naca0012.cfg file is configured such that convergence is assumed when the RMS residual in the pressure field falls below $\log_{10}(\text{RMS}) < -9$. However, you will need to change the iteration limits since we will require more than 250 iterations per run. On line 115, change the number of iterations to 99,999:

ITER= 99999

Based on the run times required for your investigation into running SU2 in parallel, select a number of processes you find suitable to run the case of the NACA 0012 airfoil until convergence, calling the warm start "restart_flow_AOA10.dat" file provided. You do not need to use the Measure-Command script in PowerShell, so if you want to monitor the convergence process you can call SU2 in parallel simply by using:

```
mpiexec -n PR "C:\SU2\v8.4.0-mpi\bin\SU2_CFD" naca0012.cfg
```

where again PR is the number of processes used in the parallel execution.

With your converged flow field data (flow.vtu), flow surface data (surface_flow.csv), and convergence history data (history.csv) obtained, rename these files to add a "_AOA10" to prevent from being overwritten when running your next case. So, for example, "flow.vtu" becomes "flow_AOA10.vtu". If you prefer, you can also simply change the output file name in the naca0012.cfg file. See lines 244, 253, and 262.

Running other cases

Repeat this process of converging your solution for $\alpha = 0$ deg and $\alpha = 15$ deg. For $\alpha = 0$ deg, you will need to uncomment line 35 of the naca0012.cfg file and comment line 38:

INC_VELOCITY_INIT= (52.1572, 0.0, 0.0)

Doing so will configure the freestream velocity to only have an x component aligned with the airfoil chordline, corresponding to $\alpha = 0$ deg. You will also need to change lines 234 and 247 to the restart_flow file associated with the $\alpha = 0$ deg case provided, "**restart_flow_AOA0**". Once the $\alpha = 0$ deg case converges, rename your flow field data, flow surface data, and convergence history data as you did with the $\alpha = 10$ deg case. Repeat this process for the $\alpha = 15$ deg case, where you will need to comment line 35 and uncomment line 41:

INC_VELOCITY_INIT= (50.37998651, 13.4992767, 0.0)

You now have the CFD-generated data needed to respond to the items required for the submission. You can find the surface pressure data in the corresponding "surface_flow.csv" files and the C_l

and C_d values in the corresponding "history.csv" files. Note that you will need to divide the surface pressure values in your surface_flow file by 0.5 to convert them to C_p.